

Polyfunctional Zinc and Magnesium Organometallics for Organic Synthesis: Some Perspectives

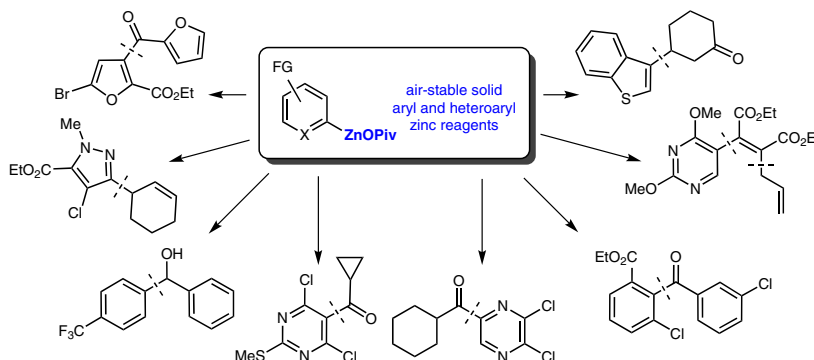
Andreas D. Benischke

Mario Ellwart

Matthias R. Becker

Paul Knochel*

Department of Chemistry, Ludwig-Maximilians-Universität München, Butenandtstraße 5-13, 81377 Munich, Germany
 Paul.Knochel@cup.uni-muenchen.de



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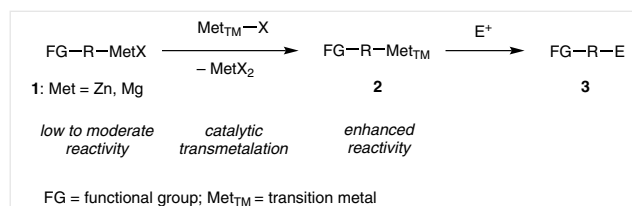
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Abstract The most important methods for the preparation of polyfunctional Zn and Mg reagents are highlighted. New perspectives for increasing the synthetic potential of these reagents including the use of Lewis acid catalysis, in situ trapping reactions, continuous flow conditions, and solid, air-stable zinc organometallics are described.

- 1 Introduction
- 2 Preparation Methods
- 3 Air-Stable Solid Organozinc Reagents
- 4 Lewis Acid Catalyst Compatibility with Polyfunctional Zinc and Magnesium Organometallics
- 5 Metalations in Flow Mode
- 6 Conclusion

Key words zinc, magnesium, lithium chloride, flow chemistry, Lewis acids

magnesium of organometallics **1** allows a range of transmetalations of **1** with transition-metal salts ($\text{Met}_{\text{TM}}\text{-X}$) leading to transition-metal intermediates **2** of palladium, copper or nickel displaying an enhanced reactivity towards various electrophiles due to the presence of d-orbitals on these transition metals (Scheme 1). This approach permits both high functional group compatibility in the reagent **1** and high reactivity towards electrophiles of the catalytic species **2** to be achieved, leading to various polyfunctional molecules **3**.



Scheme 1 Transmetalation of functionalized organometallics

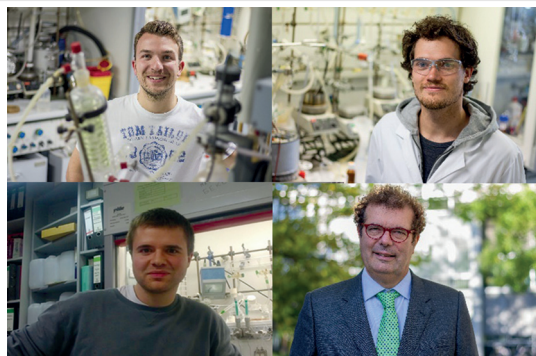
1 Introduction

Main-group organometallics such as Grignard reagents or zinc organometallics have shaped synthetic organic methodology since the infancy of organic chemistry.¹ The ever-more complex target molecules bearing multiple organic functional groups require polyfunctionalized nucleophilic reagents² for an expeditious construction of such molecules. The covalent character of the carbon–zinc bond and moderate ionicity of the carbon–magnesium bond could be exploited to prepare a variety of polyfunctional zinc and magnesium reagents bearing most functional groups (FG) encountered in organic chemistry, including functionalities with acidic C–H, O–H, or N–H bonds.³ Furthermore, the presence of two empty low-lying p-orbitals on zinc and

2 Preparation Methods

The availability of robust methods for the preparation of polyfunctional Zn- and Mg-organometallics is of high practical importance. Over the years, three methods have been shown to have particular synthetic value: first is the direct insertion of various unsaturated organic halides **4** and **5** with commercial zinc or magnesium powder in the presence of LiCl, leading to the corresponding organometallics **6** and **7** (Scheme 2).^{4,5}

This oxidative insertion has a very broad scope and the beneficial role of LiCl, which cleans the metal surface, is essential for rapid insertion. The presence of an additional Lewis acid such as ZnCl_2 (Scheme 2, eq. 2) or organoborates⁶ further catalyzes the insertion, providing numerous



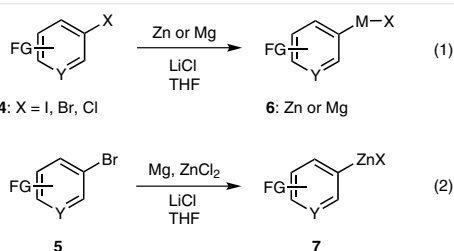
Andreas D. Benischke (upper left) was born in Burghausen (Germany) in 1989. After his undergraduate studies at the Ludwig-Maximilians-Universität München, he joined Prof. Paul Knochel's research group for his Master's thesis and started his Ph.D. in the same group in 2015. His current research work is focused on new transition-metal-catalyzed cross-coupling reactions of functionalized Mg, Mn and Zn organometallics.

Mario Ellwart (upper right) was born in Munich (Germany) in 1987. He studied chemistry at the Ludwig-Maximilians-Universität München and joined the research group of Prof. Paul Knochel in 2012. His research focuses on the synthesis of novel organozinc reagents and their applications in organic synthesis.

Matthias R. Becker (lower left) was born in Karlsruhe (Germany) in 1987. He studied chemistry at the Ludwig-Maximilians-Universität München and joined the research group of Prof. Paul Knochel in 2013. His research focuses on continuously processed metalations of unsaturated functionalized substrates.

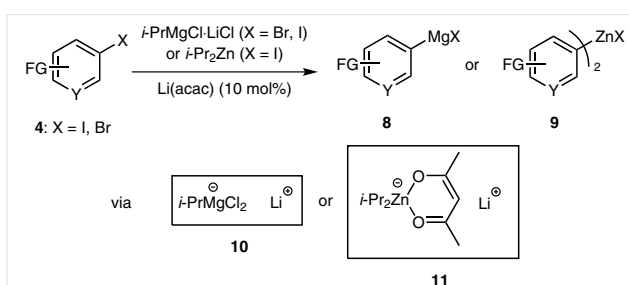
Paul Knochel (lower right) was born 1955 in Strasbourg (France). He carried out his undergraduate studies at the University of Strasbourg (France) and his Ph.D. at the ETH-Zürich with Prof. D. Seebach. He spent four years at the CNRS at the University Pierre and Marie Curie in Paris with Prof. J.-F. Normant and one year of postdoctoral studies at Princeton University in the laboratory of Prof. M. F. Semmelhack. In 1987, he accepted a position as Assistant Professor at the University of Michigan at Ann Arbor, USA. In 1991, he became a Full Professor, then in 1992, he moved to the Philipps University at Marburg (Germany) as C4-Professor in organic chemistry. In 1999, he moved to the Chemistry Department of the Ludwig-Maximilians-University in Munich (Germany). His research interests include the development of novel organometallic reagents and methods for their use in organic synthesis, asymmetric catalysis and natural product synthesis.

aryl and heteroaryl zinc reagents. A systematic study of Lewis acids⁷ including electron-transfer catalysts⁸ and the combination of Lewis acids with potential synergetic effects⁹ may further allow milder reaction conditions to be found for these insertions and thus extend further the



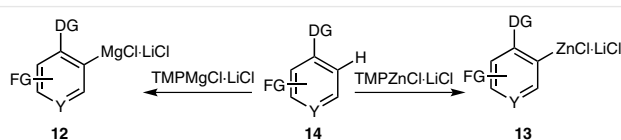
Scheme 2 LiCl-mediated Zn or Mg insertion

range of functional groups that are compatible with these insertions. The heterogeneous nature of such metal powder insertions is a drawback that led us to perform halide–magnesium or halide–zinc exchange reactions using soluble magnesium or zinc reagents (second general preparation).¹⁰ Remarkably, it was found that lithium salts (Li-Y; Y = acac, chloride) catalyze these exchange reactions,¹¹ leading to various magnesium or zinc compounds of type **8** or **9**.¹² The role of the Li-Y salts is to form ate-species¹³ of type **10** or **11** (Scheme 3), in which the nucleophilic character of the *i*-Pr moieties is increased, consequently leading to faster halide–metal exchange.



Scheme 3 Lithium salt mediated halogen–metal exchange reactions

This transient formation of ate-species such as **10** or **11** may be extended to other main-group organometallics and is related to the activation of organoboron reagents by various bases in Suzuki cross-couplings.¹⁴ The third general preparation involves directed deprotonation of polyfunctionalized unsaturated substrates. The design of reactive and selective bases for such metalations of unsaturated molecules is a very important research topic. Driven by the need to design a practical set of bases, we have envisioned the use of 2,2,6,6-tetramethylpiperidyl (TMP) bases of magnesium and zinc.¹⁵ Thus, bases such as TMPMgCl-LiCl, TMP₂Mg-2LiCl, TMPZnCl-LiCl and TMP₂Zn-2LiCl proved to be of very broad applicability^{16,17} for generating polyfunctionalized magnesium and zinc organometallics **12** or **13** starting from aromatic or heterocyclic substrates **14** (Scheme 4).

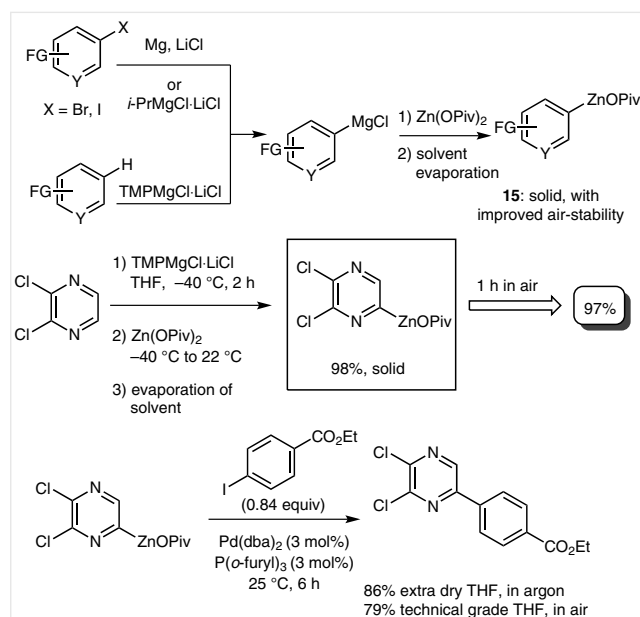


Scheme 4 Directed metalations using TMP bases

3 Air-Stable Solid Organozinc Reagents

Polyfunctional organozinc reagents readily undergo various transition-metal-catalyzed cross-couplings (e.g., Negishi cross-coupling¹⁸), however, the air- and moisture-sensitivity of zinc reagents has precluded a range of synthetic applications. Boronic esters and acids, which have

excellent air-stability, are the preferred reaction partners in Pd- or Ni-catalyzed cross-couplings.¹⁴ However, the lower reactivity and transmetalation ability of boronic derivatives as well as toxicity issues makes the availability of air-stable organozinc reagents highly desirable. We have found in our laboratories that the transmetalation of a range of organomagnesium species with zinc pivalate (OPiv = OCOTBu) leads to organozinc reagents of type **15**, which, after solvent removal, display significantly improved air- and moisture-stability (Scheme 5).¹⁹

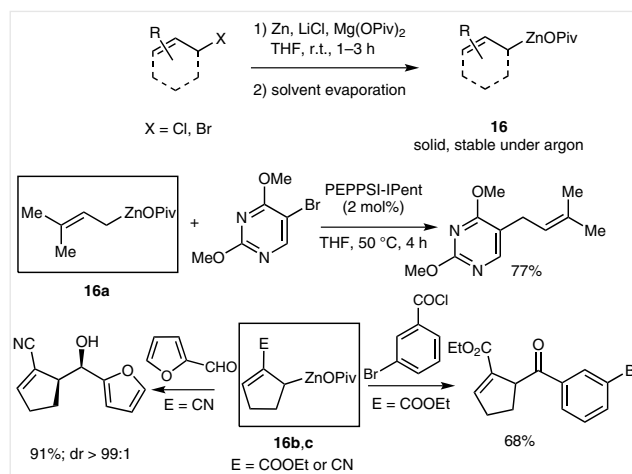


Scheme 5 Preparation of solid air-stable functionalized organozinc pivalates

These solid aryl-, heteroaryl- and benzylic zinc pivalates may find applications in high-throughput screenings. They show very similar reactivity in cross-couplings, allylations or acylations compared with standard organozinc halides. Interestingly, polyfunctional solid allylic zinc reagents such as **16a–c** have also been prepared.²⁰ Although such reagents are not air-stable, they are stable under argon for several weeks or months as solids (Scheme 6).

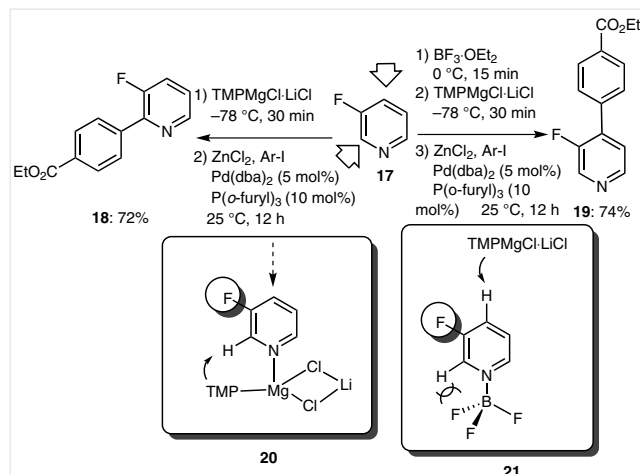
4 Lewis Acid Catalyst Compatibility with Polyfunctional Zinc and Magnesium Organo-metallics

One of the most remarkable developments of organozinc and magnesium chemistry is the discovery that these nucleophilic reagents and sterically hindered bases are compatible with strong Lewis acids.²¹ This property has led to many preparative applications. For example, it can trigger the regioselective metalation of pyridines. Thus, where-



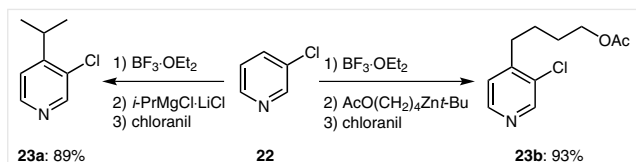
Scheme 6 Preparation of solid, substituted allylic zinc reagents and their reactions with electrophiles

as the magnesiation of 3-fluoropyridine (**17**)²¹ with TMP-MgCl·LiCl occurs in position 2, the performance of this metalation in the presence of BF₃·OEt₂ is enhanced at position 4. After a Pd-catalyzed arylation, the corresponding 2- and 4-arylated pyridines **18** and **19**, respectively, are obtained. This regioselectivity is readily explained by considering transition states such as **20** and **21** (Scheme 7).²¹



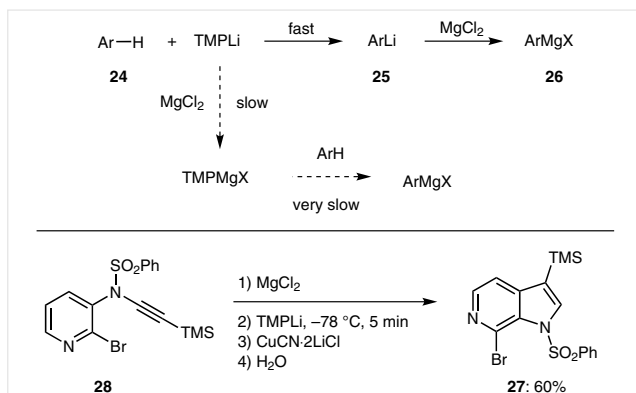
Scheme 7 Lewis acid mediated regioselective metalations

This behavior is quite general and can be extended to the magnesiation of various heterocycles.²² Furthermore, zinc and magnesium reagents proved to be compatible with BF₃·OEt₂. Thus, 3-chloropyridine **22** can be activated by BF₃·OEt₂ towards nucleophilic attack by either zinc or magnesium reagents leading to 4-substituted pyridines **23a** and **23b** (Scheme 8). Such substitutions can be extended to various heterocyclic substrates and Zn and Mg organometallics.^{23,24}



Scheme 8 Lewis acid mediated nucleophilic substitution

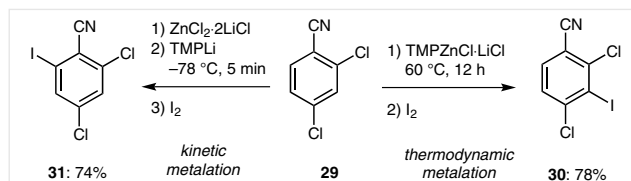
Exploring the compatibility of Lewis acids with magnesium and zinc reagents has led to preparatively valuable in situ trapping metalations²⁵ in which an unsaturated substrate **24** is mixed with a metallic salt such as MgCl_2 or ZnCl_2 followed by the addition of the strong Lewis base TMPLi. The fast metalation of Ar-H (**24**) with TMPLi produces the lithium reagent ArLi (**25**), which is then transmetalated with MgCl_2 to furnish the Grignard reagent ArMgX (**26**). Interestingly, the direct reaction of the Lewis acid MgCl_2 with TMPLi is approximately seven times slower. This in situ procedure allows a convenient synthesis of 6-azaindole **27** starting from pyridine **28** (Scheme 9).^{26a}



Scheme 9 In situ trapping metalations of functionalized aromatics

This in situ metalation proceeds well with a range of functionalized aromatics and allows regioselective magnesiations that cannot be achieved under standard conditions with $\text{TMPMgCl} \cdot \text{LiCl}$. Thus, the regular zincation of **29** with $\text{TMPZnCl} \cdot \text{LiCl}$ proceeds at the most acidic position 3 (thermodynamic metalation) leading, after iodolysis, to the 3-iodo derivative **30**. On the other hand, zincation of **29** by using in situ trapping metalation by successive treatment with $\text{ZnCl}_2 \cdot 2\text{LiCl}$ followed by TMPLi provides the 3-zincated arene (kinetic metalation) and, after iodolysis, aryl iodide **31** is obtained (Scheme 10).^{26b}

These metalations using the very strong base TMPLi have to be performed at -78°C because of the high reactivity of TMPLi. Furthermore, the scale-up of such metalations has to be optimized (adjustment of the relative amount of $\text{ZnCl}_2 \cdot 2\text{LiCl}$). These problems can be solved by performing

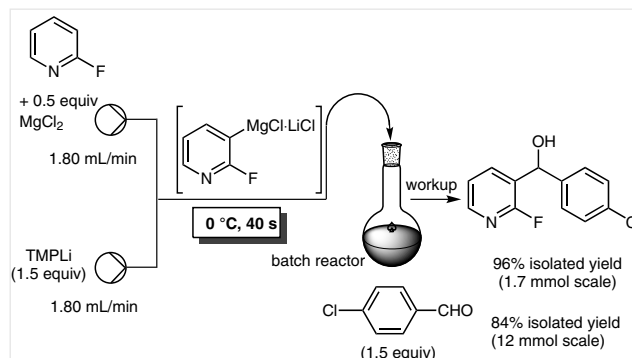


Scheme 10 Regioselectively controlled metalation of 2,4-dichlorobenzonitrile

such metalations not in a reaction vessel (batch process) but rather by conducting the same organometallic reactions in a continuous flow.

5 Metalations in Flow Mode

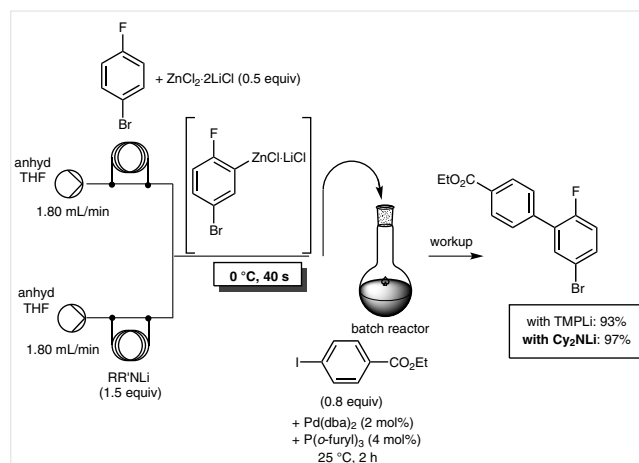
The kinetics of batch reactions differ significantly from those of the same reactions performed in continuous-flow mode.²⁷ Recently, organometallic reactions have been conducted under continuous-flow conditions, and this has led to some substantial improvements with respect to reaction temperature, safety, and scale-up.²⁸ This methodology can be applied to the in situ trapping metalation of various heterocycles using TMPLi and various salts such as $\text{ZnCl}_2 \cdot 2\text{LiCl}$, MgCl_2 or $\text{CuCl} \cdot 2\text{LiCl}$. Substantial preparative improvements have been achieved. Thus, the metalation of various electron-deficient pyridines can be achieved under continuous-flow mode at 0°C within 40 seconds, compared with a batch metalation, which requires -78°C . Moreover, the scale-up of these metalations does not require any further optimization (Scheme 11).^{29,30}



Scheme 11 Scalable in situ trapping metalations in flow mode

For prospective organometallic methods involving sensitive intermediates, the use of flow conditions pioneered by Organ, Yoshida and Ley may lead to spectacular applications in synthesis.²⁸ As a final demonstration of the power of this reaction mode, we have started to enhance the performance of metalations by using the less expensive metallic amides derived from dicyclohexylamine (Cy_2NH). This is especially attractive because the price of Cy_2NH is 100

times less than TMP-H. However, preliminary experiments have shown that $\text{Cy}_2\text{NMgCl}\cdot\text{LiCl}$, as expected, does not match the properties of $\text{TMPMgCl}\cdot\text{LiCl}$ and considerable amounts of side-products are produced in the course of the magnesiation of various aromatics with $\text{Cy}_2\text{NMgCl}\cdot\text{LiCl}$. Furthermore, the performance of flow metalations using $\text{Cy}_2\text{NMgCl}\cdot\text{LiCl}$ were not satisfactory. Nevertheless, the performance of in situ trapping flow-zincations using Cy_2NLi , which is a room-temperature-stable lithium amide, leads, like TMPLi, to excellent results in the presence of $\text{ZnCl}_2\cdot 2\text{LiCl}$ (Scheme 12).³¹



Scheme 12 In situ trapping metalations using Cy_2NLi in flow mode

6 Conclusion

It is expected that main-group organometallics will continue to play a crucial role in organic synthesis in the future. The availability of polyfunctional magnesium and zinc organometallics including solid zinc reagents with much improved air- and moisture-stability should lead to a more general acceptance of these reagents for the synthesis of complex target molecules. The compatibility of zinc and magnesium reagents with various Lewis acids should lead to numerous applications including applications in asymmetric synthesis. The ability to conduct metalations under flow conditions has already led to spectacular advances and it is expected that this industrially friendly technique will greatly extend the applications of zinc and magnesium polyfunctional organometallics. The low price and low toxicity of zinc, magnesium and lithium salts should also facilitate the handling of these organometallics on an industrial scale.

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- (30) Reactions were typically carried out in coiled reactors and tubes (1.0 mm id; 0.2–20 mL volume) made from PFA, PTFE Teflon or stainless steel.
- (31) Reactions were typically conducted on a 1.5 mmol scale. These flow reactions can be readily scaled up to a 10 mmol scale.